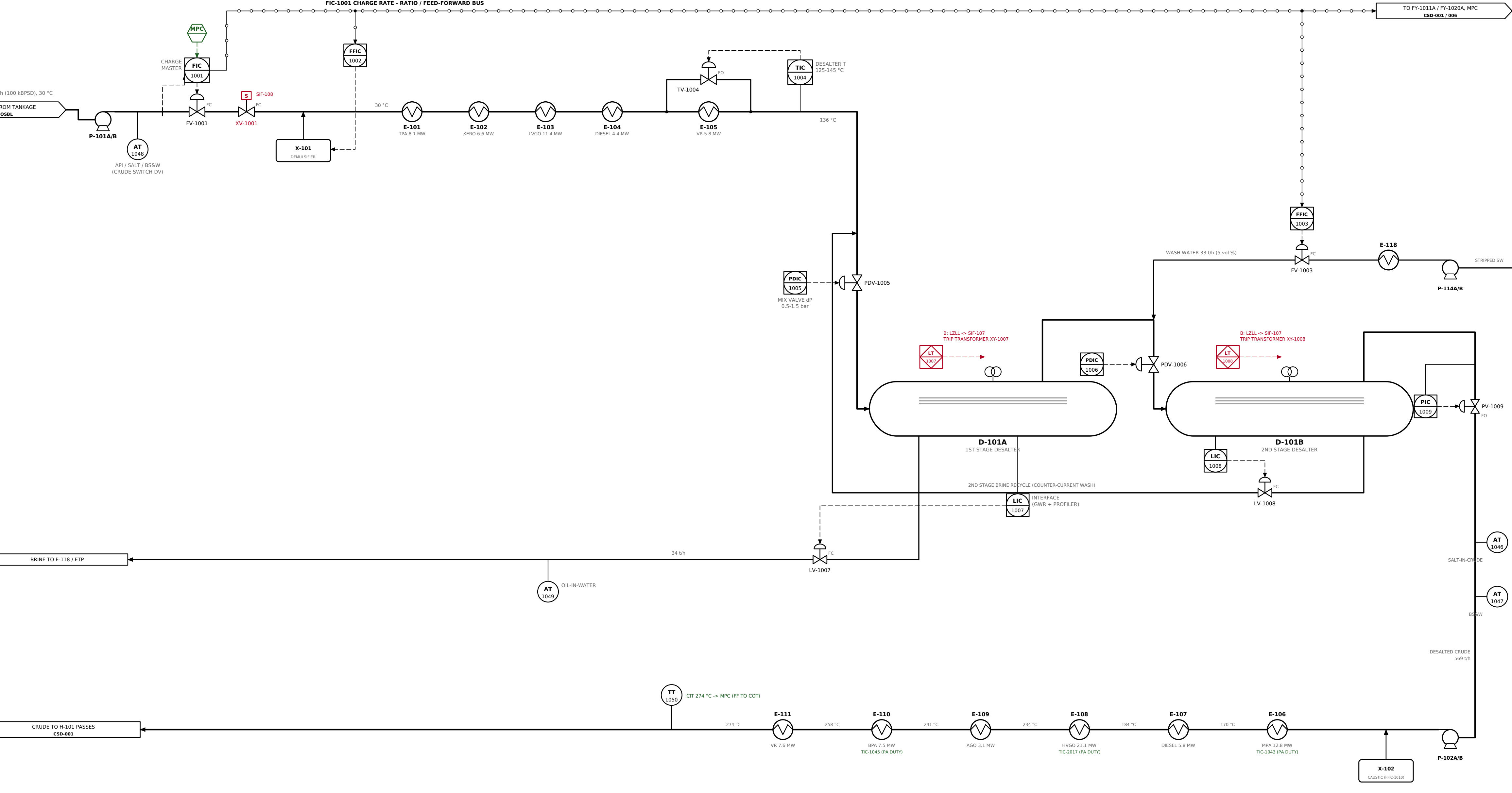




PREHEAT & DESALTING - REGULATORY LOOPS

| LOOP               | SERVICE / ACTION                                 | LIMITS       |
|--------------------|--------------------------------------------------|--------------|
| TIC-1004           | Desalter inlet T via E-105 crude bypass TV-1004  | 125-145 °C   |
| PDIC-1005/1006     | Mixing-valve dP (emulsion / salt removal)        | 0.5-1.5 bar  |
| LIC-1007/1008      | Interface level, brine LV (tight, ±5 %)          | LZLL SIF-107 |
| PIC-1009           | Desalter pressure > crude bubble point + 1.5 bar | 10.0 barg    |
| FFIC-1003          | Wash water ratio to FIC-1001                     | 4-6 vol %    |
| FFIC-1002/1010     | Demulsifier / caustic ratio to FIC-1001          | ppm          |
| TIC-1043/1045/2017 | PA return T via exchanger PA-side bypass         | MPC duty     |
| FIC-1001           | Unit throughput (MPC MV, max-feed push)          | 50-110 %     |

LEGEND (ISA-5.1 / ISA-5.4)

FIC

0000

DCS (BPCS) function, operator accessible

TC

0000

APC / MPC (L3 computer) function

Σ

Computing block: Σ sum/bias, × ratio/mult, < low select, > high select, (fx) characteriser

→

Electrical signal (4-20 mA / HART)

→

Process line

SIS

0000

SIS / BMS function (SIL rated)

FT

0000

Field-mounted instrument

↕

Control valve (FO/FC = fail open/closed)

—○—

Software / data link (DCS internal)

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APC setpoint / MPC move

CONTROL DESCRIPTION

1.

FIC-1001 is the unit throughput master (operator or MPC SPI), its PV/SP is broadcast as ratio/feed-forward bus to demulsifier, wash water, caustic, heater passes (CSD-001) and the COT feed-forward.

2.

Desalter temperature TIC-1004 (125-145 °C) positions the crude-side bypass of E-105; the VR side always flows. High temperature increases conductivity and grid current, low temperature impairs separation.

3.

Mix-valve dP PDIC-1005/1006 set by operator 0.5-1.5 bar; optimised against salt-in-crude AT-1046 and oil-in-brine AT-1049 (no closed loop on analysers).

4.

Interface LIC-1007/1008 (guided-wave radar with density profiler backup) - tight control; 2nd stage brine recycled counter-currently to 1st stage.

5.

Low-low interface SIF-107 trips the transformers.

6.

PIC-1009 holds the desalter pressure above crude vapour pressure at P-102 suction.

7.

Hot-train exchanger bypasses are on the pumparound (hot) side and belong to PA duty control (CSD-002/005). CIT is a disturbance to the H-101 COT loop (feed-forward).

8.

Crude switch: AT-1048 (density / salt) or tank-change signal triggers the MPC crude-switch feed-forward and the desalter chemical ratio presets.

| REV                                                   | DATE       | DESCRIPTION              |            |
|-------------------------------------------------------|------------|--------------------------|------------|
| A                                                     | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST                         |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT            |            |                          |            |
| PROPOSED CONTROL SCHEME                               |            |                          |            |
| CRUDE CHARGE, DESALTERS & PREHEAT TEMPERATURE CONTROL |            |                          |            |
| DISCIPLINE                                            | STAGE      | SCALE                    | SHEET      |
| INSTRUMENTATION                                       | FEED       | NTS                      | 3 OF 6     |
| DRAWN                                                 | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                           | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                        |            |                          | REV        |
| CFU-100-IC-CSD-003                                    |            |                          | A          |